

SIGNAL PROCESSING PROJECT

Seismic Data Classification

Discriminating Sensor Types using Digital Signal Processing & ML

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Project Description

🎯 Project Overview

Objective: Distinguish between 3 sensor types (Broadband, Short-Period, and Accelerometer).

Goal: Build an automated ML classification system.

Application: Digital Signal Processing for seismic pattern recognition.

🗄️ Data & Methodology

Source: Southern California Earthquake Data Center (SCEDC).

Processing: Raw seismic waveforms processed via ObsPy.

Analysis: Time-domain and Frequency-domain evaluation.

Data Strategy & Challenges



The Challenge

Initial Focus: Single event (Ridgecrest 2019).

Issue: High data imbalance.

Why? Short-Period (EHZ) sensors are rare.



The Solution

Multi-event expansion: Collected data from 4 major earthquakes (M5.1 to M7.1).

Diversity: Increased station variety across Southern California.

Tools: Python (ObsPy & Pandas) + SCEDC Database.

Sensor Selection & Final Dataset



Target Selection

Focused on High-Sample-Rate Sensors (100-200 Hz):

- **HHZ** - Broadband
- **EHZ** - Short-period
- **HNZ** - Accelerometer



Balanced Sampling Strategy

50 samples per sensor / per event:

50 Samples x 3 Sensors x 4 Events

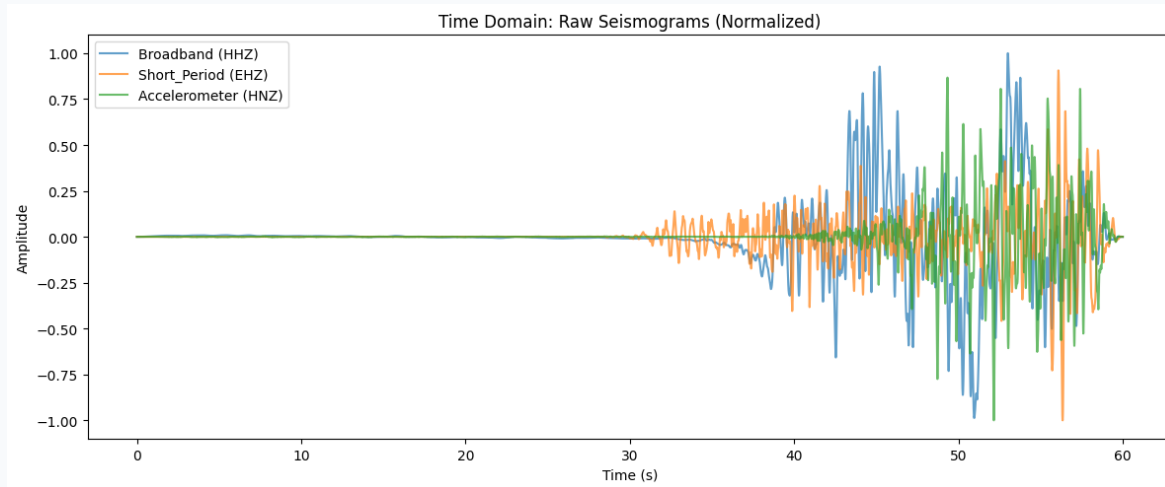
Final Dataset

600

TOTAL TRACES

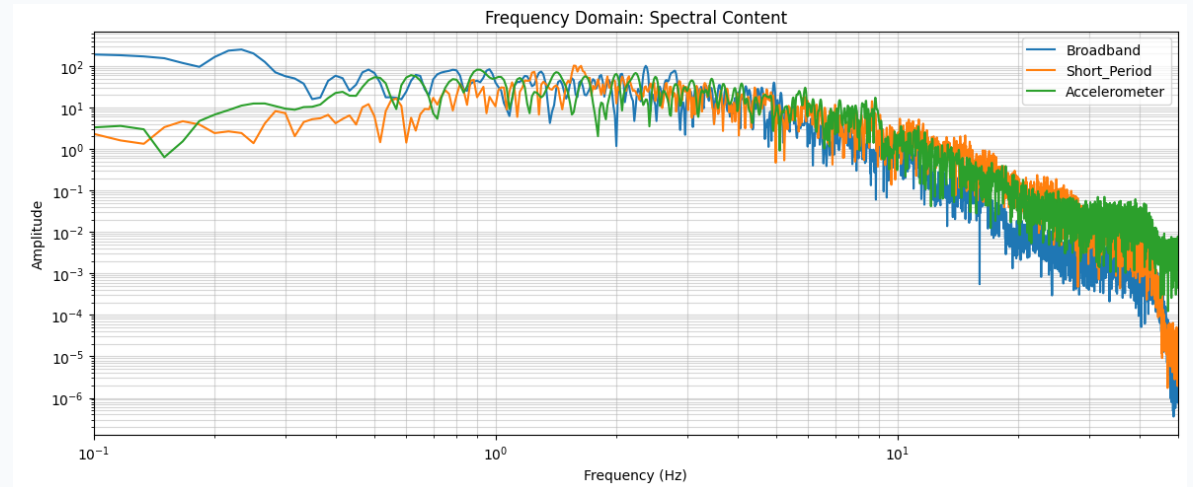
Initial Visual Analysis of a Sample

Time Domain (waveforms over time)



- **Broadband** → smooth, low-frequency oscillations
- **Short-Period** → impulsive, sharp peaks
- **Accelerometer** → medium-amplitude oscillations

Frequency Domain (waveform spectral content)



- **Broadband** → Energy below 1 Hz
- **Short-Period** → Peak 1-10 Hz
- **Accelerometer** → Flat response above 10Hz

Feature Extraction

We extracted statistical and spectral features from 100Hz resampled traces to implement the Random Forest classifier

≡ Statistical Features

- **Mean**
- **Standard Deviation**
- **Minimum**
- **Maximum**
- **Kurtosis**

Signal amplitude
and intensity

Signal shape
(smooth vs. spiky signals)

📊 Spectral Features

- **Spectral Centroid:** Center of mass of the spectrum
- **Low Energy (< 1Hz):** Distinguishes Broadband
- **High Energy (> 20Hz):** Distinguishes Accelerometers

Sensor Type Classification

RANDOM FOREST

- ✓ Robust to correlation
- ✓ Suitable for nonlinear relationships
- ✓ Measures feature importance

80 % training + 20 % testing

MODEL EVALUATION

- Accuracy
- Precision
- Recall
- F1-score

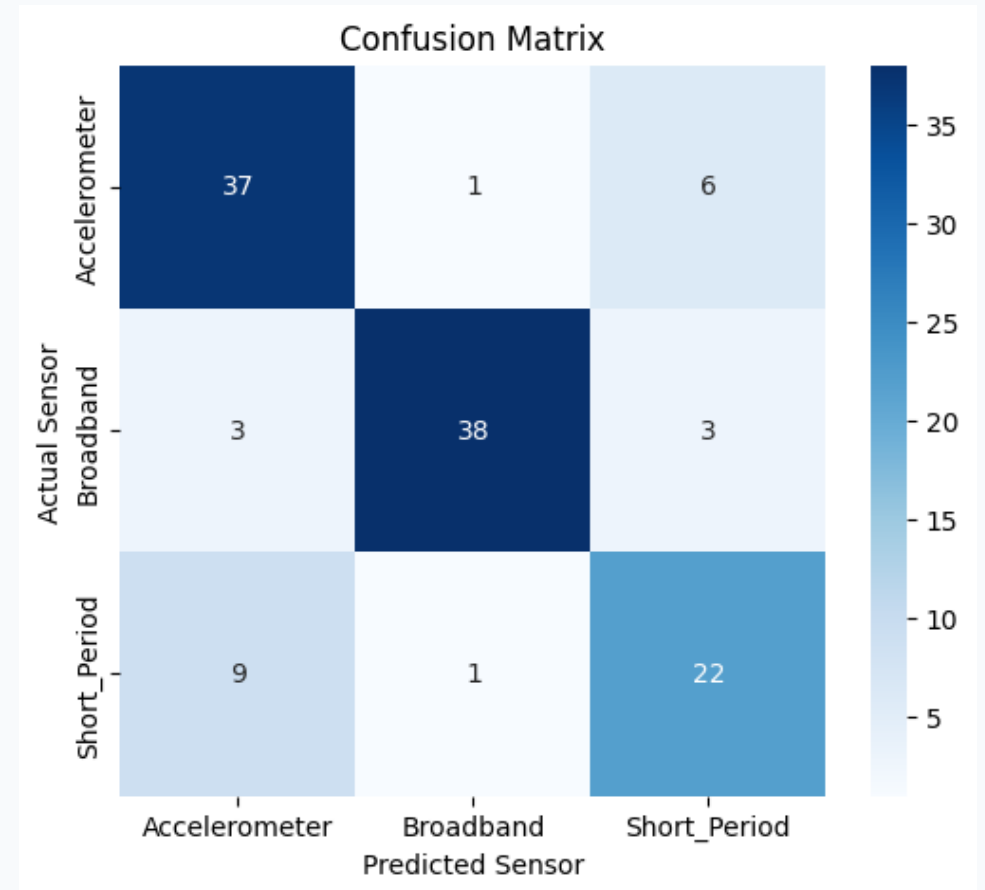
Sensor Type Classification

81%

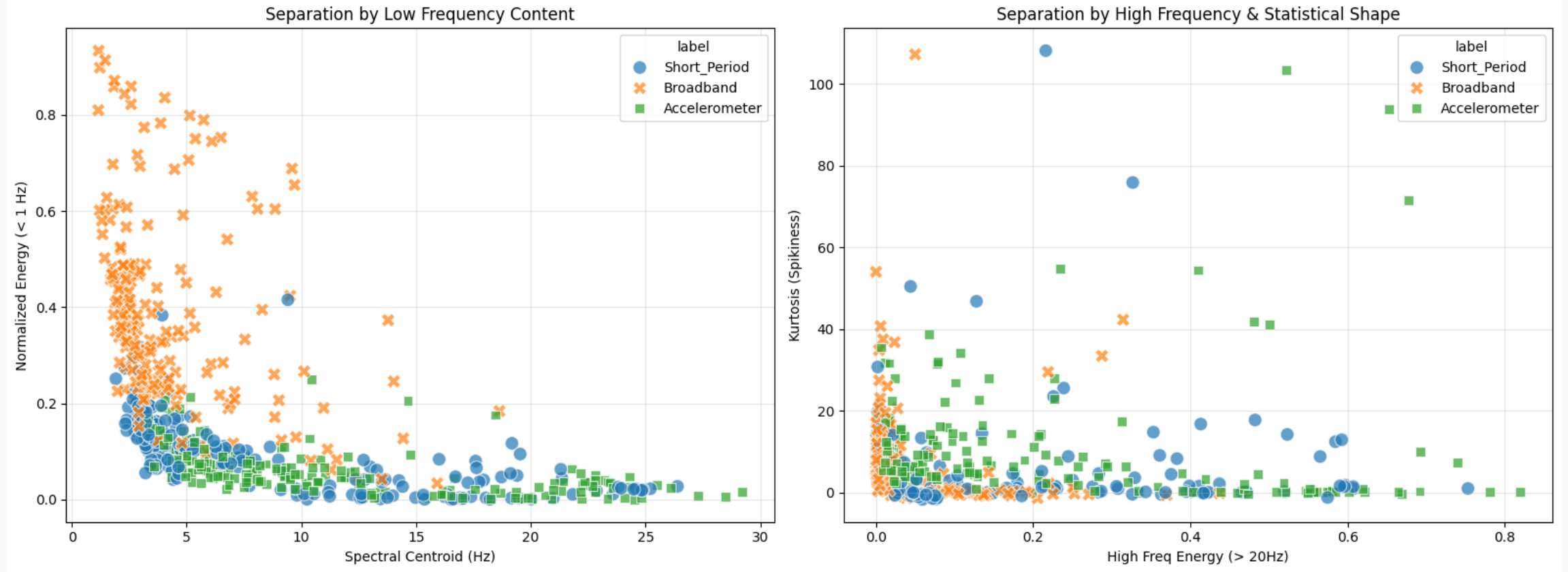
OVERALL ACCURACY

Observations

- **Broadband:** Excellent detection (F1: 0.80). Distinct low-frequency signature.
- **Short-Period:** Lowest performance (F1: 0.70)
- **Feature Importance:** Spectral Centroid and Low-Frequency Energy were top predictors.



Sensor Type Classification - visualization



Sensor-Averaged Time & Frequency Characteristics

Averaging the waveforms to analyze all samples and highlights the typical behavior of each sensor—this is the signature we want our model to learn

Methodology:

Resampling: All traces standardized to 100 Hz

Normalization: Amplitude effects removed to compare signal shape

Envelope computation: Time-domain energy evolution

FFT analysis: Frequency-domain spectral fingerprints

Average Envelope Analysis

Time-domain energy evolution averaged across all traces

Broadband (HHZ)

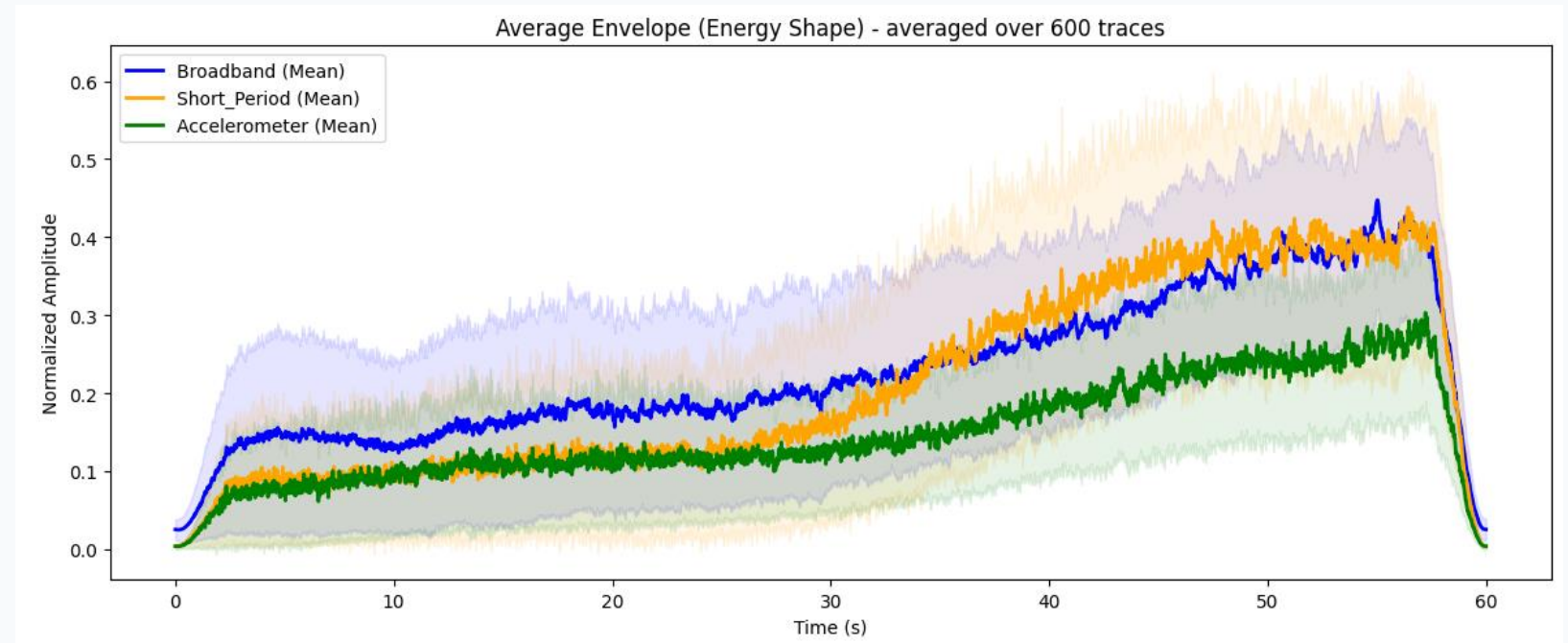
- Captures initial wave arrival best

Short-Period (EHZ)

- Dominates signal over time

Accelerometer (HNZ)

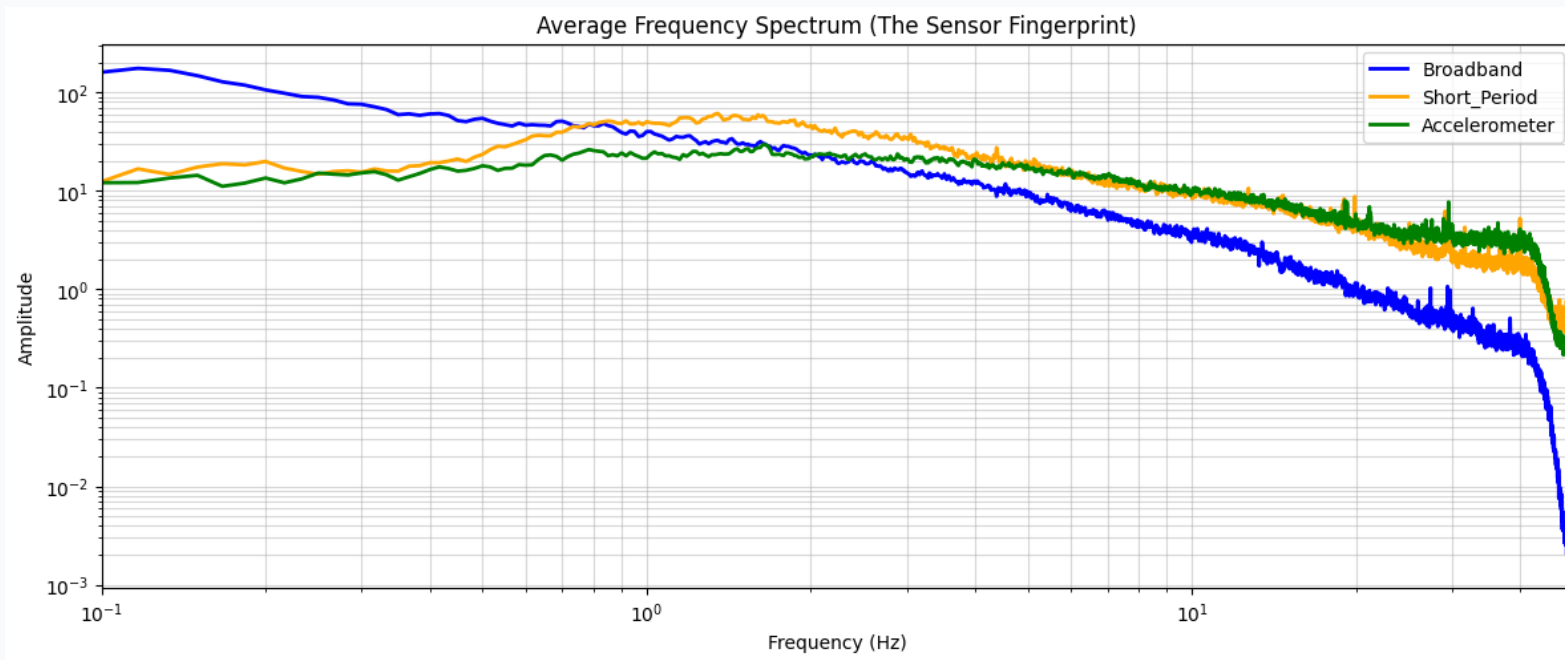
- Flat response



Time-domain patterns tell us *how quickly* each sensor reacts and how energy evolves, which hints at their design and frequency sensitivity.

Sensor Spectral Fingerprints

Averaging traces revealed distinct spectral shapes, specifically around the 1Hz mark.



The 1Hz "Cliff"

Visual inspection of the average spectra (left) revealed a critical discriminator:

- **Broadband:** Sensitive to slow waves
- **Short-Period:** Rejects low frequencies. A sharp "cliff" or drop-off in energy below 1Hz. Passband starts > 1Hz.
- **Accelerometer:** Sensitive to fast waves

Targeted Feature Extraction

To improve separation between Short-Period and Accelerometer sensors, we engineered two shape-based features focused on the 1Hz transition zone.

1. The Cliff Ratio

Ratio of energy in the passband (1-5Hz) to the rejection band (0.1-1.0Hz).

$$\text{Ratio} = E(1-5 \text{ Hz}) / E(0.1-1.0 \text{ Hz})$$

Effect: Short-Period sensors exhibit a massive ratio due to signal rejection below 1Hz.

2. Spectral Slope (1Hz)

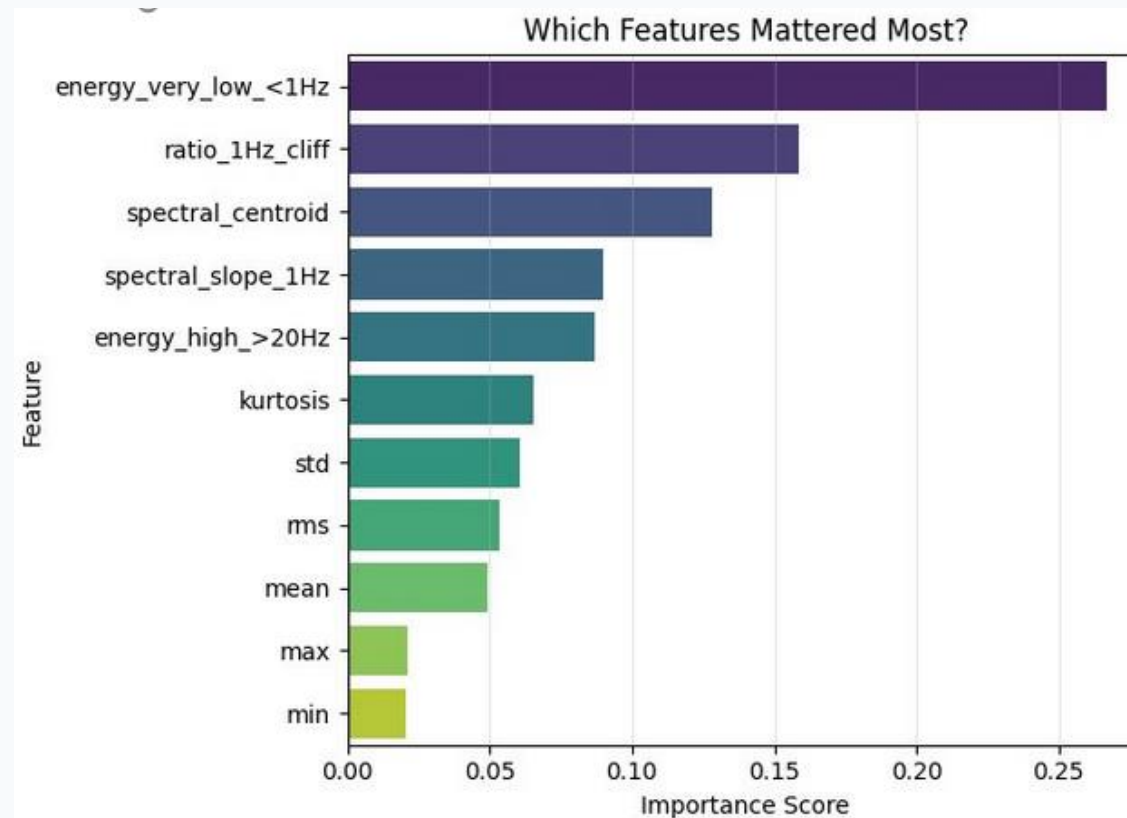
The slope of the linear regression line fitted to the log-log power spectrum between 0.5Hz and 2.0Hz.

$$\text{Slope} = \Delta \log(\text{PSD}) / \Delta \log(f)$$

Effect: Quantifies the steepness of the sensitivity onset curve.

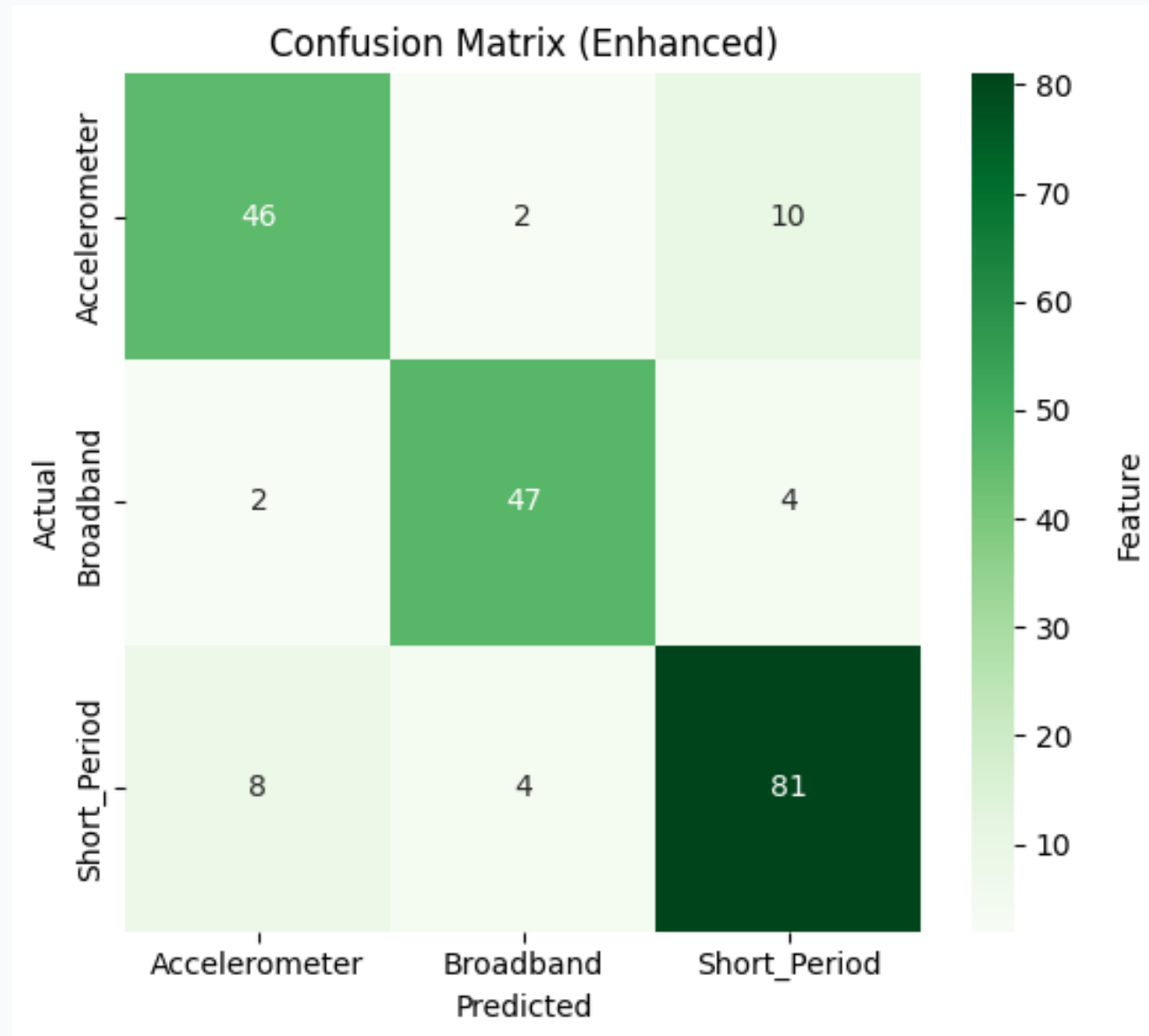
Final Model Performance

Integrating physics-informed features boosted feature importance rankings, validating the hypothesis.



The **Cliff Ratio** became the 2nd most important feature, proving that specific spectral shape descriptors are superior to generic statistics.

Improved model



Conclusion & Future Work

✔ Conclusions

- Successfully developed an ML pipeline to classify seismic sensors (HHZ, EHZ, HNZ).
- Broadband sensors are easily identified by low-frequency energy.
- Differentiation between Short-Period and Accelerometers requires specific spectral shape features (1Hz Cliff).
- Physics-informed feature engineering outperforms generic statistical features.

🚧 Future Directions

- **Streaming Data:** Validate model on continuous streams rather than pre-cut windows.
- **Deep Learning:** Utilize CNNs on spectrogram images to automate feature extraction.
- **Edge Deployment:** Deploy lightweight Random Forest models on Raspberry Pi-based seismic nodes (IoT).

